LEVEL-III

1. $\int \sqrt{1+\operatorname{cosec} x}\, dx$ **is equal to**

1) $\sin^{-1}(2\sin x+1)+c$

2) $-\sin^{-1}(2\sin x+1)+c$

3) $\sin^{-1}(2\sin x-1)+c$

4) $-\sin^{-1}(2\sin x-1)+c$

2. $\int \sqrt{\sec x-1}\, dx=$

1) $2\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

2) $\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

3) $-2\log\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

4) $2\log\left(\cos\frac{x}{2}-\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+c$

3. **If** $f(x)=\lim_{n\to\infty}\left[2x+4x^3+6x^5+\dots\dots+2nx^{2n-1}\right]$ $(0<x<1)$ **then** $\int f(x)dx$ **is equal to**

1) $-\sqrt{1-x^2}+c$ 2) $\frac{1}{\sqrt{1-x^2}}+c$

3) $\frac{1}{x^2-1}+c$ 4) $\frac{1}{1-x^2}+c$

4. $\int \frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx$ **is equal to**

1) $-\sin 2x-\sin x+c$

2) $-\frac{1}{2}\sin 2x+\sin x+c$

3) $-\frac{1}{2}\sin 2x-\sin x+c$

4) $\frac{1}{2}\sin 2x+\sin x+c$

5. **If** $\alpha>1$, **then** $\int \frac{dx}{x^2+2\alpha x+1}=$

1) $\frac{1}{\sqrt{1-\alpha^2}}\tan^{-1}\left(\frac{x+\alpha}{\sqrt{1-\alpha^2}}\right)+c$

2) $\frac{1}{\sqrt{1-\alpha^2}}\cot^{-1}\left(\frac{x+\alpha}{\sqrt{1-\alpha^2}}\right)+c$

3) $\frac{1}{2\sqrt{\alpha^2-1}}\log\left(\frac{x+\alpha-\sqrt{\alpha^2-1}}{x+\alpha+\sqrt{\alpha^2-1}}\right)+c$

4) $\frac{1}{2\sqrt{\alpha^2-1}}\log\left(\frac{x+\alpha+\sqrt{\alpha^2-1}}{x+\alpha-\sqrt{\alpha^2-1}}\right)+c$

6. $\int \frac{\sec^2 x dx}{(\sec x+\tan x)^{10}}\, dx=$

1) $\frac{-1}{9}(\sec x+\tan x)^{-9}-\frac{1}{11}(\sec x+\tan x)^{-11}+c$

2) $\frac{-1}{2}\left[\frac{-1}{9}(\sec x+\tan x)^{-9}-\frac{1}{11}(\sec x+\tan x)^{-11}\right]+c$

3) $\frac{1}{2}\left[\frac{-1}{9}(\sec x+\tan x)^{-9}-\frac{1}{11}(\sec x+\tan x)^{-11}\right]+c$

4) $\frac{1}{2}\left[-\frac{1}{9}(\sec x+\tan x)^{-9}+\frac{1}{11}(\sec x+\tan x)^{-11}\right]+c$

7. **Evaluate** $\int \frac{x^2(x\sec^2 x+\tan x)}{(x\tan x+1)^2}dx$

1) $\frac{x^2}{x\tan x+1}+2\log|x\sin x+\cos x|+C$

2) $\frac{-x^2}{x\tan x+1}+2\log|x\sin x+\cos x|+C$

3) $\frac{-x^2}{x\cot x+1}+2\log|x\sin x+\cos x|+C$

4) $\frac{-x^2}{x\tan x+1}+2\log|x\cos x+\cos x|+C$

8 $\int \frac{1}{(2x-7)\sqrt{x^2-7x+12}}dx$ **is equal to**

1) $2\sec^{-1}(2x-7)+c$

2) $\sec^{-1}(2x-7)+c$

3) $\frac{1}{2}\sec^{-1}(2x-7)+c$

4) $4\sec^{-1}(2x-7)+c$

9. **If** $\int \frac{e^x-e^{-x}}{e^{2x}+e^{-2x}}dx = A$ **log** $\left|\frac{e^x+e^{-x}+a}{e^x+e^{-x}-a}\right|+c$,

then $(A, a)=$

1) $\left(\frac{-1}{2\sqrt{2}}, -\sqrt{2}\right)$ 2) $\left(\frac{-1}{\sqrt{2}}, 2\sqrt{2}\right)$

3) $\left(\frac{1}{\sqrt{2}}, -2\sqrt{2}\right)$ 4) $\left(\frac{1}{2\sqrt{2}}, -\sqrt{2}\right)$

10. $\int \frac{\log(x+1)-\log x}{x(x+1)}dx=$

1) $\log(x-1)\log x+\frac{1}{2}(\log(x-1))^2-\frac{1}{2}(\log x)^2+c$

2) $\frac{1}{2}(\log(x+1))^2+\frac{1}{2}(\log x)^2-\log(x+1)\log x+c$

3) $-\frac{1}{2}(\log(x+1))^2-\frac{1}{2}(\log x)^2+\log x.\log(x+1)+c$

4) $\left[\log\left(1+\frac{1}{x}\right)\right]^2+c$

11. $\int \frac{1}{\sec x+\tan x+\operatorname{cosec} x+\cot x}dx=$

1) $\frac{1}{2}(\sin x+\cos x+x)+k$

2) $\frac{1}{2}(\sin x-\cos x-x)+k$

3) $\frac{1}{2}(\sin x+\cos x-x)+k$

4) $\frac{1}{2}(-\sin x+\cos x-x)+k$

12. $\int \sqrt{e^{2x}-1}\,dx=$

1) $\sqrt{e^{2x}-1}-\sec^{-1}(e^x)+c$

2) $\sqrt{e^{2x}-1}+\sec^{-1}(e^x)+c$

3) $\frac{1}{2}\sqrt{e^{2x}-1}-\sec^{-1}(e^x)+c$

4) $2\sqrt{e^{2x}-1}-\sec^{-1}(e^x)+c$

13. $\int \mathrm{Tan}(x-\alpha)\mathrm{Tan}(x+\alpha)\mathrm{Tan}2x\,dx$

1) $\log\left|\frac{\sqrt{\sec 2x}}{\sec(x+\alpha)\sec(x-\alpha)}\right|+c$

2) $\log\left|\frac{\sec 2x}{\sec(x+\alpha)\sec(x-\alpha)}\right|+c$

3) $\log\left|\frac{2\sqrt{\sec 2x}}{\sec(x+\alpha)\sec(x-\alpha)}\right|+c$

4) $\log\left|\frac{\sqrt{\sec x}}{\sec(x+\alpha)\sec(x-\alpha)}\right|+c$

14. $\int \frac{x^2}{(x\,Cos\,x-Sin\,x)^2}$ **dx =**

1) cot x + $\frac{x\,Co\sec x}{x\,Cos\,x-Sin\,x}+c$

2) $-\left[\cot x+\frac{x\,Co\sec x}{x\,Cos\,x-Sin\,x}\right]+c$

3) - cot x + $\frac{x\,Co\sec x}{x\,Cos\,x-Sin\,x}+c$

4) cot x - $\frac{x\,Co\sec x}{x\,Cos\,x-Sin\,x}$ + c

15. **If** $\int \frac{1}{(x\tan x+1)^2}dx = k\left(\frac{\tan x}{x\tan x+1}\right)+c$ **then**

$k=$

1) -1 2) 1

3) 2 4) $\frac{1}{2}$

16. $\int \frac{e^{\text{Sin}\,x}\left(x\ \text{Cos}^3 x - \text{Sin}\ x\right)}{\text{Cos}^2\ x}\,dx =$

1) $e^{Sin\,x}\left(x - Sec\,x\right)+c$
2) $e^{Sin\,x}\left(x + Sec\,x\right)+c$
3) $e^{Sin\,x}\left(x\,Cos\,x + Sin\,x\right)+c$
4) $e^{Sin\,x}\left(x\,Cos\,x - Sin\,x\right)+c$

17. $\int (\sqrt{\cot x}+\sqrt{\tan x})dx =$

1) $\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2}\tan x}\right)+c$
2) $2\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2}\tan x}\right)+c$
3) $\sqrt{2}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2\tan x}}\right)+c$
4) $\sin^{-1}\left(\sin x - \cos x\right)+c$

18. $\int \frac{1}{\cos^6 x + \sin^6 x}dx =$

1) $Tan^{-1}(\tan x - \cot x) + c$
2) $Cot^{-1}(\tan x - \cot x) + c$
3) $Sin^{-1}(\tan x - \cot x) + c$
4) $\cos^{-1}(\tan x - \cot x) + c$

19. If $\int f(x)\sin x\cos x\,dx = \frac{\log|f(x)|}{2(b^2-a^2)}+c,$ **then f(x) can be**

1) $\frac{1}{a^2\sin^2 x + b^2\cos^2 x}$
2) $\frac{1}{a^2\sin^2 x - b^2\cos^2 x}$
3) $\frac{1}{a^2\cos^2 x + b^2\sin^2 x}$
4) $\frac{1}{a^2\cos^2 x - b^2\sin^2 x}$

20. $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}}dx$ **is equal to**

1) $\tan^{-1}\sqrt{\frac{x^2+x+1}{x}}+c$
2) $2\tan^{-1}\sqrt{\frac{x^2+x+1}{x}}+c$
3) $3\tan^{-1}\sqrt{\frac{x^2+x+1}{x}}+c$
4) $4\tan^{-1}\sqrt{\frac{x^2+x+1}{x}}+c$

21. If $\int \frac{(\sin\theta-\cos\theta)}{(\sin\theta+\cos\theta)\sqrt{\sin\theta\cos\theta+\sin^2\theta\cos^2\theta}}d\theta$, $=\operatorname{cosec}^{-1}\left[f(\theta)\right]+c$ **then**

1) $f(\theta)=\sin 2\theta+1$ 2) $f(\theta)=1-\sin 2\theta$
3) $f(\theta)=\sin 2\theta-1$ 4) $f(\theta)=\sin\theta+1$

22. $\int \frac{\sin^3 x}{\left(1+\cos^2 x\right)\sqrt{1+\cos^2 x+\cos^4 x}}dx =$

1) $\sec^{-1}\left(\sec x+\cos x\right)+c$
2) $\sec^{-1}\left(\sec x-\cos x\right)+c$
3) $\sec^{-1}\left(\sec x-\tan x\right)+c$
4) $\sec^{-1}\left(\sec x+\tan x\right)+c$

23. The value of $\int \frac{x\cos x+1}{\sqrt{2x^3e^{\sin x}+x^2}}dx$ **is**

1) $\ln\left|\frac{\sqrt{2xe^{\sin x}+1}-1}{\sqrt{2xe^{\sin x}+1}+1}\right|+c$
2) $\ln\left|\frac{\sqrt{2xe^{\sin x}-1}-1}{\sqrt{2xe^{\sin x}-1}+1}\right|+c$
3) $\ln\left|\frac{\sqrt{2xe^{\sin x}-1}+1}{\sqrt{2xe^{\sin x}-1}-1}\right|+c$
4) $\ln\left|\frac{\sqrt{2xe^{\sin x}+1}+1}{\sqrt{2xe^{\sin x}-1}+1}\right|+c$

24. $\int \frac{\sec x}{\sqrt{\sin(2x+\alpha)+\sin\alpha}}dx$ **is equal to**

1) $\sqrt{2\sec\alpha(\tan x+\tan\alpha)}+c$

2) $\sqrt{2\sec\alpha(\tan x-\tan\alpha)}+c$

3) $\sqrt{2\sec\alpha(\tan\alpha-\tan x)}+c$

4) $\sqrt{2\tan\alpha(\sec x-\sec\alpha)}+c$

25. $\int\left(\frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}\right)dx=$

1) $\sin x - \frac{2}{\sin x} - 6\,\mathrm{Tan}^{-1}(\sin x)+c$

2) $\sin x + \frac{2}{\sin x} - 6\,\mathrm{Tan}^{-1}(\sin x)+c$

3) $\sin x - \frac{2}{\sin x} + 6\,\mathrm{Tan}^{-1}(\sin x)+c$

4) $\sin x + \frac{2}{\sin x} + 6\,\mathrm{Tan}^{-1}(\sin x)+c$

26. $\int \frac{dx}{1-\sin^4 x}=$

1) $\frac{1}{2}[\mathrm{Tan}\,x - \frac{1}{\sqrt{2}}\mathrm{Tan}^{-1}(\sqrt{2}\,\mathrm{Tan}\,x)]+c$

2) $\frac{1}{2}[\mathrm{Tan}\,x + \frac{1}{\sqrt{2}}\mathrm{Tan}^{-1}(\sqrt{2}\,\mathrm{Tan}\,x)]+c$

3) $\frac{1}{2}[\mathrm{Tan}\,x + \frac{1}{2\sqrt{2}}\cot^{-1}(\sqrt{2}\,\mathrm{Tan}\,x)]+c$

4) $\frac{1}{2}[\mathrm{Tan}\,x - \frac{1}{2\sqrt{2}}\cot^{-1}(\sqrt{2}\,\mathrm{Tan}\,x)]+c$

27. $\int \frac{1}{\sin^3 x\cos^5 x}dx=$

1) $3\log Tan\,x+\frac{3}{2}\tan^2 x+\frac{1}{4}\tan^4 x-\frac{1}{2}\cot^2 x+c$

2) $3\log\tan x+\frac{3}{2}\tan^2 x+\frac{1}{4}\tan^4 x-\frac{1}{2}\tan^2 x+c$

3) $3\log\cot x+\frac{3}{2}\cot^2 x+\frac{1}{4}\cot^4 x-\frac{1}{2}\tan^2 x+c$

4) $3\log\cot x+\frac{3}{2}\cot^2 x+\frac{1}{4}\cot^4 x-\frac{1}{2}\cot^2 x+c$

28. If $\int\frac{dx}{(2\sin x+\sec x)^4}=Pt^{-5}+Qt^{-6}+Rt^{-7}$ **,**

where $t=1+\tan x$**, then** $P+Q+R=$

1) $\frac{16}{105}$ 2) $-\frac{16}{105}$ 3) $-\frac{1}{35}$ 4) $\frac{1}{35}$

29. The value of $\int\frac{x^5+x^4+4x^3+4x^2+4x+4}{(x^2+2)^3}dx$ **is equal to**

1) $\frac{4x-3}{(x^2+1)^2}+\frac{3}{8}\frac{x}{x^2+2}+\frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+c$

2) $\frac{1}{12}\frac{2x-3}{(x^2+1)^2}+\frac{3}{16}\frac{x}{x^2+2}+\frac{3}{16\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+c$

3) $\frac{2x-3}{(x^2+1)^2}+\frac{3}{8}\frac{x}{x^2+2}+\frac{1}{2\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+c$

4) $\frac{1}{2}\log|x^2+2|+\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right)+c$

30. If the curve f(x) = $\int e^x$ dx passes through (0, 1) then the ascending order of f(0), f(1), f(-1), f(2) is

1) f(0), f(1), f(-1), f(2)
2) f(-1), f(0), f(1), f(2)
3) f(2), f(1), f(0), f(-1)
4) f(1), f(-1), f(2), f(0)

31. Let $f(x)$ **be a function such that** $f(0)=f^1(0)=0$ **and** $f^{11}(x)=\sec^4 x+4$ **then** $f(x)=$

1) $\log(\sin x)+\frac{1}{3}\tan^3 x+6x$

2) $\frac{2}{3}\log(\sec x)+\frac{1}{6}\tan^2 x+2x^2$

3) $\log(\cos x)+\frac{1}{6}\cos^2 x+\frac{x^2}{5}$

4) $\log(\sec x)+\frac{1}{6}\sin^3 x+x^2$

32 .If $x\in\left(\pi,\frac{3\pi}{2}\right)$ **then**

$\int\left(\sqrt{1+\sin x}-\sqrt{1-\sin x}\right)dx=$

1) $4\cos\frac{x}{2}+c$ 2) $-4\cos\frac{x}{2}+c$

3) $4\sin\frac{x}{2}+c$ 4) $-4\sin\frac{x}{2}+c$

33. $\int\sqrt{x+\sqrt{x^2+2}}\;dx=$

1) $\left(\frac{x+\sqrt{x^2+2}}{3}\right)^{5/2}-2\left(x+\sqrt{x^4+2}\right)^{-3/2}+c$

2) $\frac{2}{7}\left(x+\sqrt{x^2+2}\right)^{7/2}-2\left(x+\sqrt{x^2+2}\right)+c$

3) $\frac{\left(x+\sqrt{x^2+2}\right)^{3/2}}{3}-2\left(x+\sqrt{x^2+2}\right)^{-1/2}+c$

4) $\frac{\left(x+\sqrt{x^2+2}\right)^{9/2}}{3}+2\left(x+\sqrt{x^2+2}\right)^{-5/2}+c$

34. $\int\frac{1}{(x+1)^2\sqrt{x^2+2x+2}}dx=$

1) $-\frac{\sqrt{x^2+2x+2}}{x+1}+c$

2) $\frac{\sqrt{x^2+2x+2}}{x+1}+c$

3) $2\frac{\sqrt{x^2+2x+2}}{x+1}+c$

4) $\frac{1}{2}\frac{\sqrt{x^2+2x+2}}{x+1}+c$

35. If $f(x)$ **is a primitive of** $\frac{\tan x-\sin x}{x^3}(x\neq 0)$

then the value of $\underset{x\to 0}{Lt}\,f'(x)=$

1) 2 2) $\frac{1}{2}$ 3)1 4) 3

36. $\int\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2}\cdot\frac{dx}{x}=$

1) $-2\left[\log\left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right)-\cos^{-1}\sqrt{x}\right]+c$

2) $-2\left[\log\left(\frac{1-\sqrt{1-x}}{\sqrt{x}}\right)+\sin^{-1}\sqrt{x}\right]+c$

3) $-2\left[\log\left(\frac{1-\sqrt{1-x}}{\sqrt{x}}\right)-\sin^{-1}\sqrt{x}\right]+c$

4) $\log\left(\frac{1+\sqrt{1-x}}{\sqrt{x}}\right)+\cos^{-1}\sqrt{x}+c$

37. $\int\frac{dx}{\sin x+\sec x}=$

1) $\frac{1}{2\sqrt{3}}\log\left|\frac{\sin x-\cos x+\sqrt{3}}{\cos x-\sin x+\sqrt{3}}\right|+\tan^{-1}(\sin x+\cos x)+c$

2) $\frac{1}{2\sqrt{3}}\log\left|\frac{\sin x-\cos x-\sqrt{3}}{\sin x-\cos x+\sqrt{3}}\right|+c$

3) $\frac{1}{2\sqrt{3}}\log\left|\left(\sin x-\cos x+\sqrt{3}\right)\left(\sin x-\cos x-\sqrt{3}\right)\right|+c$

4) $\frac{1}{2\sqrt{3}}\log\left|\left(\sin x-\cos x+\sqrt{3}\right)\left(\sin x-\cos x-\sqrt{3}\right)\right|$
$+\tan^{-1}(\sin x+\cos x)+c$

38. If

$\int\frac{dx}{(\sin x+4)(\sin x-1)}=\frac{A}{\tan\frac{x}{2}-1}+B\tan^{-1}f(x)+c$ *then*

1) $A=\frac{1}{5},B=\frac{2}{5\sqrt{15}},f(x)=\frac{4\tan x+3}{\sqrt{15}}$

2) $A=-\frac{1}{5},B=\frac{2}{\sqrt{15}},f(x)=\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}$

3) $A=\frac{2}{5},B=-\frac{2}{5\sqrt{15}},f(x)=\frac{4\tan x+1}{\sqrt{15}}$

4) $A=\frac{2}{5},B=-\frac{2}{5\sqrt{15}},f(x)=\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}$

39. **If** $f(x)=\lim_{n\to\infty} n^2\left(x^{1/n}-x^{1/(n+1)}\right), x>0$ **then** $\int x f(x)dx$ **is equal to**

1) $x^2/2+c$ 2) 0

3) $x^2\log x-\frac{1}{2}x^2+c$ 4) $\frac{x^2}{2}\log x-\frac{x^2}{4}+c$

40. **Let** $f(x)$ **be a polynomial of degree three such that** $f(0)=1, f(1)=2$ **and x=0 is critical point of** $f(x)$ **such that** $f(x)$ **does not have a local extremum at x=0 then** $\int\frac{f(x)}{x^2+1}dx$ **is equal to**

1) $x-\log\left(x^2+1\right)+\tan^{-1}x+c$

2) $x+(1/2)\log\left(x^2+1\right)-\tan^{-1}x+c$

3) $(1/2)x^2+(1/2)\log\left(x^2+1\right)-\tan^{-1}x+c$

4) $(1/2)\left(x^2-\log\left(x^2+1\right)\right)+\tan^{-1}x+c$

41. $\int\frac{x\,dx}{\sqrt[2012]{\left(1+x^2\right)^{1012}\left(2+x^2\right)^{3012}}}=\frac{\alpha}{2\beta}\left(1-f(x)\right)^{\beta/\alpha}+c$

1) $f\left(\sqrt{2}\right)=\frac{1}{\beta-\alpha}$ 2) $f\left(\sqrt{2}\right)=\frac{1}{\alpha-2\beta}$

3) $f(1)=\frac{1}{\alpha-2\beta}$ 4) $f\left(\sqrt{3}\right)=\frac{1}{\alpha-\beta}$

42. **If**

$$\int\frac{(x\cos\alpha+1)}{\left(x^2+2x\cos\alpha+1\right)^{3/2}}dx=\frac{f(x)}{\sqrt{x^2+2x\cos\alpha+1}}+c$$

then $f(x)=$

1) x 2) $2x$

3) $3x$ 4) x^2

43. $\int\frac{3x^2+2x}{x^6+2x^5+x^4+2x^3+2x^2+5}dx=$

1) $\frac{1}{4}\tan^{-1}\left(\frac{x^3+x^2+1}{2}\right)+c$

2) $\frac{1}{2}\tan^{-1}\left(\frac{x^3+x^2+1}{2}\right)+c$

3) $\sin^{-1}\left(\frac{x^3+x^2+1}{2}\right)+c$

4) $\frac{1}{2}\tan^{-1}\left(\frac{x^3+x^2}{2}\right)+c$

44. **If** $\int(\sin 3\theta+\sin\theta)\cos\theta\, e^{\sin\theta}d\theta=$

$\left(A\sin^3\theta+B\cos^2\theta+C\sin\theta+D\cos\theta+E\right)e^{\sin\theta}+F$

then D=

1) 4 2) 0 3) 12 4) -12

45. $\int\frac{e^{\sqrt{x}}}{\sqrt{x}}\left(x+\sqrt{x}\right)dx$ **equals**

1) $2e^{\sqrt{x}}\left[x-\sqrt{x}+1\right]+C$ 2) $e^{\sqrt{x}}\left[x-2\sqrt{x}+1\right]$

3) $e^{\sqrt{x}}\left[x+\sqrt{x}\right]+C$ 4) $e^{\sqrt{x}}\left[x+\sqrt{x}+1\right]+C$

46. $\int\frac{e^{\tan^{-1}x}}{\left(1+x^2\right)}\left[\left(\sec^{-1}\sqrt{1+x^2}\right)^2+\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]dx$ $(x>0)$

1) $e^{\tan^{-1}x}.\tan^{-1}x+C$ 2) $\frac{e^{\tan^{-1}x}.\left(\tan^{-1}x\right)^2}{2}+C$

3) $e^{\tan^{-1}x}.\left(\sec^{-1}\left(\sqrt{1+x^2}\right)\right)^2+C$

4) $e^{\tan^{-1}x}.\left(\operatorname{cosec}^{-1}\left(\sqrt{1+x^2}\right)\right)^2+C$

47. **Let** $f(x)=\frac{2\sin^2x-1}{\cos x}+\frac{\cos x(2\sin x+1)}{1+\sin x}$ **then** $\int e^x\left(f(x)+f'(x)\right)dx$ **equals (where c is the constant of integration)**

1) $e^x\tan x+c$ 2) $e^x\cot x+c$

3) $e^x\operatorname{cosec}^2x+c$ 4) $e^x\operatorname{cosec}x+c$

48. If y is implicit differentiable function of x such that $y(x+y)^2 = x$ then $\int \frac{dx}{x+3y} =$

1) $ln\left((x+y)^2+1\right)+c$

2) $\frac{1}{2}ln\left((x-y)^2+1\right)+c$

3) $ln\left((x-y)^2+1\right)+c$

4) $\frac{1}{2}ln\left((x+y)^2+1\right)+c$

49. A curve $g(x)=\int x^{27}\left(1+x+x^2\right)^6\left(6x^2+5x+4\right)dx$ is passing through (0,0) then

1) $g(1)=\frac{3^7}{7}$ 2) $g(1)=\frac{2^7}{7}$

3) $g(2)=\frac{1}{7}$ 4) $g(-1)=\frac{3^7}{14}$

50. $\int \frac{e^{\cot x}}{\sin^2 x}\left(2\ln \operatorname{cosec} x+\sin 2x\right)dx =$

1) $-2e^{\cot x}\ln(\operatorname{cosec} x)+c$ 2) $e^{\cot x}\ln x+c$

3) $e^{\cot x}\ln(\operatorname{cosec} x)+c$ 4) $e^{\cot x}\ln(\sin x)+c$

51. $\int\left(\sqrt{\frac{\cos x}{x}}-\sqrt{\frac{x}{\cos x}}\sin x\right)dx$ **equals**

1) $-\sqrt{x\cos x}+C$ 2) $\sqrt{x\sin x}+C$

3) $2\sqrt{x\cos x}+C$ 4) $C-2\sqrt{x\cos x}$

52. The value of $\int(\sin 101x).\sin^{99}x\,dx$ is $\frac{\sin(100x)\sin^{100}x}{k+5}$, then $\frac{k}{19}$ is

1) 5 2) 4 3) -4 4) -2

53. I f $\int \frac{x+2}{\sqrt{4x-x^2}}dx =$

$-\sqrt{4x-x^2}+k\sin^{-1}\frac{x-2}{2}+c$ **then k =**

1) 1 2) 2 3) 3 4) 4

54. If $I=\int \frac{\sin 2x}{(3+4\cos x)^3}dx$, then I is equal to

1) $\frac{3\cos x+8}{(3+4\cos x)^2}+C$ 2) $\frac{3+8\cos x}{16(3+4\cos x)^2}+C$

3) $\frac{3+\cos x}{(3+4\cos x)^2}+C$ 4) $\frac{3-8\cos x}{16(3+4\cos x)^2}+C$

55. If the graph of the antiderivative $F(x)$ of $f(x)=\log(\log x)+(\log x)^{-2}$ passes through (e, 2014-e) then the term independent of x in F (x) is 'K', then sum of digits of 'K' is

1) 5 2) 6 3) 7 4) 8

56. If $\int \frac{\sqrt[3]{\left(1+2x^3\right)^2}}{x^6}dx=-\frac{1}{K}\left(L+\frac{1}{x^3}\right)^{\frac{M}{N}}+c$ (HCF of M and N is 1). The value of (K+L+M+N)

1) 5 2) 10 3) 15 4) 20

57. If $f\left(\frac{3x-4}{3x+4}\right)=x+2$ then $\int f(x)dx=$

1) $e^{x+2}\log\left|\frac{3x-4}{3x+4}\right|$

2) $\frac{-8}{3}\log|1-x|+\frac{2}{3}x+c$

3) $\frac{8}{3}\log|x-1|+\frac{x}{3}+c$ 4) $\frac{3x-4}{3x+4}+c$

58. $\int e^x\left(\frac{2\tan x}{1+\tan x}+\cot^2\left(x+\frac{\pi}{4}\right)\right)dx$ **is eqal to**

1) $e^x\tan\left(\frac{\pi}{4}-x\right)+c$ 2) $e^x\tan\left(x-\frac{\pi}{4}\right)+c$

3) $e^x\tan\left(\frac{3\pi}{4}-x\right)+c$ 4) $e^x\tan\left(\frac{6\pi}{4}-x\right)+c$

59. The value of $\int\left(x^2+x\right)\left(x^{-8}+2x^{-9}\right)^{\frac{1}{10}}dx$ is

1) $\frac{5}{11}\left(x^2+2x\right)^{\frac{11}{10}}+c$ 2) $\frac{5}{6}(x+1)^{\frac{11}{10}}+c$

3) $\frac{6}{7}(x+1)^{11/10}+c$ 4) $\frac{3}{5}(x+1)^{11/10}+c$

60. $\int e^{x^4}\left(x+x^3+2x^5\right)e^{x^2}dx$ **is euqal to**

1) $\frac{1}{2}xe^{x^2}e^{x^4}+c$ 2) $\frac{1}{2}x^2e^{x^4}+c$

3) $\frac{1}{2}e^{x^2}e^{x^4}+c$ 4) $\frac{1}{2}x^2e^{x^2}e^{x^4}+c$

61. $\int x.\frac{l\,n\left(x+\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}dx=$

$a\sqrt{1+x^2}\,l\,n\left(x+\sqrt{1+x^2}\right)+bx+c$, **then**

1) $a=1,\ b=-1$ 2) $a=1,\ b=1$

3) $a=-1,\ b=1$ 3) $a=-1,\ b=-1$

62. Let $f(x)=\frac{x}{\left(1+x^n\right)^{\frac{1}{n}}}$ **for** $n\geq 2$ **and** $g(x)=\underbrace{fofo........of}_{f\ occurs\ n\ times}(x)$ **. then** $\int x^{n-2}g(x)dx$ **equals to**

1) $\frac{1}{n(n-1)}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

2) $\frac{1}{n-1}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

3) $\frac{1}{n(n-1)}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

4) $\frac{1}{n+1}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

63. $\int\frac{2\sin 2\phi-\cos\phi}{6-\cos^2\phi-4\sin\phi}d\phi=$

1) $\log\left|\sin^2\phi-4\sin\phi+5\right|+7\tan^{-1}\left(\sin\phi-2\right)+c$

2) $2\log\left|\sin^2\phi-4\sin\phi+5\right|+7\tan^{-1}\left(\sin\phi-2\right)+c$

3) $2\log\left|\sin^2\phi-4\sin\phi+5\right|$

4) $\log\left|\sin^2\phi+4\sin\phi+5\right|+7\tan^{-1}\left(\sin\phi-2\right)+c$

LEVEL - III - KEY

1)3	2)3	3)4	4)3	5)3	6)3
7)2	8)2	9)4	10)3	11)2	12)1
13)1	14)3	15)2	16)1	17)3	18)1
19)1	20)2	21)1	22)1	23)1	24)1
25)1	26)2	27)1	28)2	29)4	30)2
31)2	32)3	33)3	34)1	35)2	36)1
37)1	38)4	39)4	40)4	41) 3	42) 1
43) 2	44) 2	45) 1	46) 3	47) 1	48) 4
49) 1	50) 1	51) 3	52) 1	53) 4	54) 2
55) 3	56) 3	57) 2	58) 2	59) 1	60) 4
61) 1	62) 1	63) 2			

LEVEL - III - HINTS

1. $I=\int\sqrt{\frac{1+\sin x}{\sin x}}dx$

$Put\ \sin x=t\Rightarrow dx=\frac{dt}{\sqrt{1-t^2}}$

$\therefore I=\int\frac{1}{\sqrt{t-t^2}}dt=\int\frac{dt}{\sqrt{\left(\frac{1}{2}\right)^2-\left(t-\frac{1}{2}\right)^2}}$

2. Put $\cos\frac{x}{2}=t$

3. $f(x)=\underset{n\to\infty}{lt}\ 2x\left(1+2x^2+3x^4+.....n.x^{2n-2}\right)$

we get $f(x)=\frac{2x}{(1-x^2)^2}\ as\ 0<x<1$

4. Multiplying Nr and Dr by $\sin 3x$ and using transformations.

5. $x^2+2\alpha x+\alpha^2+1-\alpha^2=\left(x+\alpha\right)^2-\left(\sqrt{\alpha^2-1}\right)^2$

6. Put sec x + tan x = t so that sec x - tan x = 1/t

$\Rightarrow\sec x=\frac{t^2+1}{2t}$

7. we have

$I=x^2\times\frac{-1}{x\tan x+1}-2\int x\frac{-1}{(x\tan x+1)}dx$

$\frac{-x^2}{x\tan x+1}+2\log\left|x\sin x+\cos x\right|+C$

8. $\int \frac{2dx}{(2x-7)\sqrt{4x^2-28x+48}}$

$= \int \frac{2dx}{(2x-7)\sqrt{(2x-7)^2-1}}$

9. $I = \int \frac{d\left(e^x+e^{-x}\right)}{\left(e^x+e^{-x}\right)^2-2} = \frac{1}{2\sqrt{2}}\ln\left|\frac{e^x+e^{-x}-\sqrt{2}}{e^x+e^{-x}+\sqrt{2}}\right|+c$

10. Put $1+\frac{1}{x}=t$

11. since

$(\sin x+\cos x+1)(\sin x+\cos x-1)=2\sin x\cos x$

$\frac{\sin x\cos x}{1+\sin x+\cos x}=\frac{1}{2}(\sin x+\cos x-1)$

12. $I=\int \frac{e^{2x}-1}{\sqrt{e^{2x}-1}}dx=\frac{1}{2}\int\frac{2e^{2x}}{\sqrt{e^{2x}-1}}-\int\frac{e^x}{e^x\sqrt{\left(e^x\right)^2-1}}dx$

13. $\tan(x-\alpha)\tan(x+\alpha)\tan 2x$

$=\tan 2x-\tan(x+\alpha)-\tan(x-\alpha)$

14. $-\int \frac{-x\sin x}{(x\cos x-\sin x)^2}\frac{x}{\sin x}dx$ and use integration by parts

15. $I=\int\frac{\cos^2 x}{(x\sin x+\cos x)^2}dx=\int\underbrace{\frac{x\cos x}{(x\sin x+\cos x)^2}}_{V}\underbrace{\left(\frac{\cos x}{x}\right)}_{U}dx$

16. $\int \underbrace{x}_{u}\,\underbrace{e^{\sin x}\cos x\,dx}_{v}-\int \underbrace{e^{\sin x}}_{u}\,\underbrace{\sec x\tan x\,dx}_{v}$

17. Put $\tan x=t^2$ $\Rightarrow GI=2\int\frac{1+t^2}{1+t^4}dt$

(or) $GI=\int\frac{(\sin x+\cos x)}{\sqrt{\sin x\cos x}}dx$

$=\sqrt{2}\int\frac{(\sin x+\cos x)}{\sqrt{1-(\sin x-\cos x)^2}}dx$

Put $\sin x-\cos x=t$

18. $I=\int\frac{\sec^6 x}{1+\tan^6 x}dx$ put $\tan x=t$

19. Diff. both sides w.r.t. x

$f(x).\sin x.\cos x=\frac{f'(x)}{2\left(b^2-a^2\right)f(x)}$

$-b^2\int\cos x(-\sin x)dx-a^2\int\sin x.\cos x dx$

$=\frac{1}{2}\int\frac{f'(x)}{\left(f(x)\right)^2}dx$

$\Rightarrow\frac{-b^2}{2}\cos^2 x-\frac{a^2}{2}\sin^2 x=\frac{-1}{2f(x)}$

20. $G.I=\int\frac{x^2-1}{(x^2+2x+1).x\sqrt{x+\frac{1}{x}+1}}dx$

put $x+\frac{1}{x}+1=t^2$.

21. Multiplying Nr and Dr by $2(\cos\theta+\sin\theta)$

$I=\int\frac{-2(\cos^2\theta-\sin^2\theta)}{(\cos\theta+\sin\theta)^2\sqrt{4\sin^2\theta\cos^2\theta+4\sin\theta\cos\theta}}d\theta$

$=\int\frac{-2\cos 2\theta}{(1+\sin 2\theta)\sqrt{(1+\sin 2\theta)^2-1}}d\theta$

and put $1+\sin 2\theta=t$

22. $\therefore I=\int\frac{\sin^3 x\,dx}{\cos^2 x(\sec x+\cos x)\sqrt{(\sec x+\cos x)^2-1}}$

and Put $\sec x+\cos x=t$

23. $\therefore G.I=\int\frac{(\cos x+\frac{1}{x})dx}{\sqrt{2x.e^{\sin x}+1}}$ Put $2x.e^{\sin x}+1=t^2$.

24. $I=\int\frac{\sec x\,dx}{\sqrt{2\sin(x+\alpha)\cos x}}$

$=\frac{1}{\sqrt{2}}\int\frac{\sec^2 x}{\sqrt{\tan x.\cos\alpha+\sin\alpha}}dx$.

Put $\tan x.\cos\alpha+\sin\alpha=t^2$

25. Put $\sin x=t$

26. $1-\sin^4 x=\cos^2 x(1+\sin^2 x)$ & then divide Numerator & Denominator by $\cos^4 x$

27. Put $\tan x=t$

28. G.I $=\int \frac{\sec^4 x}{\left(2\tan x+\sec^2 x\right)^4}dx$

$=\int \frac{\left(1+\tan^2 x\right)\sec^2 x}{\left(1+\tan x\right)^8}dx$ Put $1+\tan x=t$

29. $x^5+x^4+4x^3+4x^2+4x+4=(x^2+2)^2(x+1)$

30. $f^1(x)=e^x>0 \quad \forall x\in R$

$\Rightarrow f(x)$ is increasing on R

$\therefore f(-1)<f(0)<f(1)<f(2)$

31. $f^1(x)=\tan x+\frac{\tan^3 x}{3}+4x+c_1$

$f^1(0)=0\Rightarrow c_1=0$. Integrating once again

32. $\frac{x}{2}\in\left(\frac{\pi}{2},\frac{3\pi}{4}\right)\Rightarrow c+s>0$ and $c-s<0$

$\sqrt{1+\sin x}-\sqrt{1-\sin x}=2\cos\left(\frac{x}{2}\right)$

33. Put $x+\sqrt{x^2+2}=t \quad x^2+2=(t-x)^2$

$2tx=t^2-2 \quad x=\frac{1}{2}\left(t-\frac{2}{t}\right)$

34. $I=\int\frac{1}{(x+1)^2\sqrt{(x+1)^2+1}}dx$

Put $x+1=\frac{1}{t}$ then $I=-\int\frac{t}{\sqrt{t^2+1}}dt$

35. $f(x)=\int\left(\frac{\tan x-\sin x}{x^3}\right)dx$

$\Rightarrow f'(x)=\frac{\tan x-\sin x}{x^3}=\frac{\tan x(1-\cos x)}{x^3}$

36. Put $x=\cos^2\theta$

$GI=\int\left(\frac{1-\cos\theta}{1+\cos\theta}\right)^{\frac{1}{2}}\left(\frac{-2\sin\theta\cos\theta}{\cos^2\theta}\right)d\theta$

$=-2\int\frac{2\sin^2\left(\frac{\theta}{2}\right)}{\cos\theta}d\theta=-2\int\left(\frac{1-\cos\theta}{\cos\theta}\right)d\theta$

$=-2\left(\log\left|\sec\theta+\tan\theta\right|-\theta\right)+c$

37. $I=\int\frac{2\cos x\,dx}{2\sin x\cos x+2}$

$=\int\frac{(\cos x+\sin x)}{2+\sin 2x}dx+\int\frac{(\cos x-\sin x)}{2+\sin 2x}dx$

$=\int\frac{\cos x+\sin x}{3-(\sin x-\cos x)^2}dx+\int\frac{(\cos x-\sin x)}{1+(\sin x+\cos x)^2}dx$

38. $I=\frac{1}{5}\int\left(\frac{1}{(\sin x-1)}-\frac{1}{(\sin x+4)}\right)dx$ and put $\tan\frac{x}{2}=t$.

39. $f(x)=\underset{n\to\infty}{lt}\, n^2.x^{\frac{1}{n+1}}\left(x^{\frac{1}{n}-\frac{1}{n+1}}-1\right)$

$=\underset{n\to\infty}{lt}\frac{x^{\frac{1}{n+1}}}{(1+\frac{1}{n})}.\underset{n\to\infty}{lt}\left(\frac{x^{\frac{1}{n(n+1)}}-1}{\frac{1}{n(n+1)}}\right)=\frac{x^0}{1}.\log x$.

40. Let $f(x)=ax^3+bx^2+cx+d$ using $f(0)=1,f(1)=2$ and at $x=0$, $f^1(x)=0$, $f^{11}(x)=0 \Rightarrow f(x)=x^3+1$

41. Put $\left(\frac{1+x^2}{2+x^2}\right)=t$ then we get

$\alpha=503,\beta=250,f(x)=\frac{1}{2+x^2}$

42. $\int\frac{x^3\left(\frac{\cos\alpha}{x^2}+\frac{1}{x^3}\right)}{x^3\left(1+\frac{2\cos\alpha}{x}+\frac{1}{x^2}\right)^{3/2}}$

put $1+\frac{2\cos\alpha}{x}+\frac{1}{x^2}=t$

43. Given integral $=\int\frac{3x^2+2x}{\left(x^3+x^2+1\right)^2+4}dx$

Put $x^3+x^2+1=t$

$= \int \frac{1}{t^2+4} dt = \frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right) + c$

$= \frac{1}{2}\tan^{-1}\left(\frac{x^3+x^2+1}{2}\right) + c$

44. Put $\sin\theta = t$ we get

$\int (4t - 4t^3)e^t dt = \begin{pmatrix} At^3 + B(1-t^2) \\ +Ct + D\sqrt{1-t^2} + E \end{pmatrix}$

$e^t + F$

D=0

Differentiating both sides wrt to t, we get B=-12

45. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\left(x+\sqrt{x}\right) dx$; put $x = t^2$; $dx = 2t\, dt$

$= \int e^t (t^2+t)\, dt = e^t (At^2 + Bt + C)$ (Let)

Diffrentiate both the sides

$e^t (t^2 + t) = e^t (2At + B) + (At^2 + Bt + C) e^t$

On comparing coefficient we get

A = 1 ; B = – 1 ; C = 1]

46. note that $\sec^{-1}\sqrt{1+x^2} = \tan^{-1}x$;

$\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\tan^{-1}x$ for $x > 0$

$I = \int \frac{e^{\tan^{-1}x}}{1+x^2}\left((\tan^{-1}x)^2 + 2\tan^{-1}x\right) dx$

put $\tan^{-1}x = t$ $= \int e^t (t^2+2t)\, dt$

$= e^t . t^2 = e^{\tan^{-1}x}\left(\tan^{-1}x\right)^2 + C$

47. $\frac{\cos x(1+2\sin x)}{1+\sin x} - \frac{\cos^2 x - \sin^2 x}{\cos x}$

$= \frac{\cos^2 x(1+2\sin x) - (1+\sin x)(\cos^2 x - \sin^2 x)}{\cos x(1+\sin x)}$

$= \frac{-\sin x\cos^2 x + \sin^3 x}{\cos x(1+\sin x)}$

$= \frac{\sin x\cos^2 x + \sin^2 x(1+\sin x)}{\cos x(1+\sin x)}$

$= \frac{\sin x(1-\sin x) + \sin^2 x}{\cos x} = \tan x$

48. $y(x+y)^2 = x \Rightarrow \frac{dy}{dx} = \frac{1-2y(x+y)}{(x+y)(x+3y)}$

$\Rightarrow 1 + \frac{dy}{dx} = \frac{(x+y)^2+1}{(x+y)(x+3y)}$

$\frac{(x+y)(dx+dy)}{(x+y)^2} = \frac{dx}{x+3y}$

$\therefore \int \frac{dx}{x+3y} = \frac{1}{2}\ln\left((x+y)^2+1\right) + c$

49. $g(x) = \int x^{24}.(1+x+x^2)^6.(6x^5+5x^4+4x^3)dx$

$= \int (x^4+x^5+x^6)^6.(6x^5+5x^4+4x^3)dx$

Put $t = x^6+x^5+x^4 = \int t^6 dt = \frac{t^7}{7} + C$

$= \frac{(x^6+x^5+x^4)^7}{7} + C$

But it passes through (0,0) $\Rightarrow C = 0$

$\therefore g(1) = 3^7/7$

50. $t = \cot x \Rightarrow -\operatorname{cosec}^2 x dx = dt \Rightarrow dx = \frac{-1}{1+t^2}dt$

$I = -\int e^t\left(\ln(1+t^2) + \frac{2t}{1+t^2}\right)dt$

$= -e^t \ln(1+t^2) + c = -2e^{\cot x}\ln \operatorname{cosec} x + c$

51. $\int \left(\sqrt{\frac{\cos x}{x}} - \sqrt{\frac{x}{\cos x}}\sin x\right) dx = \int \frac{\cos x - x\sin x}{\sqrt{x\cos x}} dx$

$= 2\sqrt{x\cos x}$

52. $I = \int (\sin 100x.\cos x + \cos 100x.\sin x)\sin^{99} x.dx$

$I = \int \underset{I}{\sin 100x.\cos x}.\underset{II}{\sin^{99} x}dx$

$+\int \cos 100x.\sin^{100} x.dx$

$I = \frac{\sin(100x)\sin^{100} x}{100} - \frac{100}{100}\int \cos(100x)\sin^{100} x dx$

$+\int \cos(100x).\sin^{100} x.dx$

$I = \frac{\sin(100x).\sin^{100} x}{100} + C$

53. $I=\int \frac{x+2}{\sqrt{(4x-x^2)}}dx = \frac{\int -\frac{1}{2}(4-2x)+4}{\sqrt{4x-x^2}}dx$

$=-\frac{1}{2}\int\frac{(4-2x)}{\sqrt{4x-x^2}}dx+4\int\frac{dx}{\sqrt{4-(x-2)^2}}$

(put $4x-x^2=t$) $=-\frac{1}{2}\int\frac{dt}{\sqrt{t}}+4\int\frac{dx}{\sqrt{2^2-(x-2)^2}}$

$=-\frac{1}{2}\times 2\sqrt{t}+4\sin^{-1}\left(\frac{x-2}{2}\right)$

54. Write $I=\int\frac{2\sin x\cos x}{(3+4\cos x)^3}dx$ and put $3+4\cos x=t$ and so that $-4\sin x\,dx=dt$ and

$I=\frac{-1}{8}\int\frac{(t-3)}{t^3}dt=\frac{1}{8}\left(\frac{1}{t}-\frac{3}{2}\frac{1}{t^2}\right)+C$

$=\frac{2t-3}{16t^2}=\frac{8\cos x+3}{16(3+4\cos x)^2}+C$

55. An antidevivative of

$f(x)=F(x)=\int \log(\log x)+(\log x)^{-2}dx+C$ =

$x log(\log x)-x.(log x)^{-1}+C$ Put

$x=e, we\,have\,2014-e=e.0+e+c\Rightarrow C=2014.$

56. m =-6; n =3; p= 2/3 in $\int x^m\left(a+bx^n\right)^p dx$

$\frac{m+1}{n}+p=\frac{-5}{3}+\frac{2}{3}=p=-1$ (integer)

Put $1+2x^3=x^3t^3$ ($\because$ put $a+bx^n=x^nt^k$),

$t^3=x^{-3}+2$, $3t^2dt=-3x^{-4}dx$,

$dx=-x^4t^2dt$

$\therefore -\int x^{-6}\left(x^3t^3\right)^{2/3}x^4t^2dt$

$-\int t^4dt=-\frac{t^5}{5}+c=-\frac{1}{5}\left(2+\frac{1}{x^3}\right)^{5/3}+c$

$\therefore$ K+L +M +N =5+2+5+3=15

57. Let $\frac{3x-4}{3x+4}=t\ \Rightarrow x=\frac{4t+4}{3-3t}$

58. $\int e^x\left(\frac{2\tan x}{1+\tan x}+\tan^2\left(x-\frac{\pi}{4}\right)\right)dx$

$=\int e^x\left(\tan\left(x-\frac{\pi}{4}\right)+\sec^2\left(x-\frac{\pi}{4}\right)\right)dx$

$=e^x\tan\left(x-\frac{\pi}{4}\right)+c$

59. Given that $\int\left(x^2+x\right)\left(x^{-8}+2x^{-9}\right)^{\frac{1}{10}}dx$ or

$\int(x+1)\left(x^2+2x\right)^{\frac{1}{10}}dx$

now put $x^2+2x=t\Rightarrow(x+1)dx=\frac{dt}{2}$

$\Rightarrow\int t^{\frac{1}{10}}.\frac{dt}{2}=\frac{1}{2}\times\frac{10}{11}t^{\frac{11}{10}}=\frac{5}{11}t^{\frac{11}{10}}+c$

$=\frac{5}{11}\left(x^2+2x\right)^{\frac{11}{10}}+c$

60. Putting $x^2=t$ $I=\frac{1}{2}\int e^{t^2}\left(1+t+2t^2\right)e^tdt$

$=\frac{1}{2}\int e^t\left[te^{t^2}+\left(e^{t^2}+2t^2e^{t^2}\right)\right]dt$

$=\frac{1}{2}\int e^t\left[f(t)+t'(t)\right]dt=\frac{1}{2}e^t\left(te^{t^2}\right)+c$

where $t=x^2$

61. $I=\int x\frac{\ln\left(x+\sqrt{x^2+1}\right)}{\sqrt{x^2+1}}dx$, let $\sqrt{x^2+1}=t$

$\Rightarrow\frac{dt}{dx}=\frac{x}{\sqrt{x^2+1}}$

$\Rightarrow I=\int\ln\left(t+\sqrt{t^2-1}\right)dt$

$=\ln\left(t+\sqrt{t^2-1}\right).t-\int\frac{1+\frac{t}{\sqrt{t^2-1}}}{t+\sqrt{t^2-1}}$

$$= t.\ln\left(t+\sqrt{t^2-1}\right)-\frac{1}{2}\int\frac{2t}{\sqrt{t^2-1}}dt$$

$$= t.\ln\left(t+\sqrt{t^2-1}\right)-\sqrt{t^2-1}+C$$

$$= \sqrt{1+x^2}.\ln\left(x+\sqrt{1+x^2}\right)-x+C$$

$$\Rightarrow a=1,\ b=-1$$

62. We have $fof(x)=\dfrac{f(x)}{\left[1+f(x)^n\right]^{\frac{1}{n}}}=\dfrac{x}{\left(1+2x^n\right)^{\frac{1}{n}}}$

Also, $fofof(x)=\dfrac{x}{\left(1+3x^n\right)^{\frac{1}{n}}}$

$\therefore\ g(x)=\underbrace{(fofo....of)}_{n\ terms}(x)=\dfrac{x}{\left(1+nx^n\right)^{\frac{1}{n}}}$

Thus, $\int x^{n-2}g(x)\,dx=\int\dfrac{x^{n-1}dx}{\left(1+nx^n\right)^{\frac{1}{n}}}$

$$=\frac{1}{n^2}\int\frac{n^2x^{n-1}dx}{\left(1+nx^n\right)^{\frac{1}{n}}}=\frac{1}{n^2}\int\frac{\frac{d}{dx}\left(1+nx^n\right)}{\left(1+nx^n\right)^{\frac{1}{n}}}$$

$$=\frac{1}{n(n-1)}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$$

63. $I=\displaystyle\int\frac{2\sin 2\phi-\cos\phi}{6-\cos^2\phi-4\sin\phi}d\phi$

$$=\int\frac{(4\sin\phi-1)\cos\phi}{\sin^2\phi-4\sin\phi+5}d\phi$$

$I=\displaystyle\int\frac{4t-1}{t^2-4t+5}dt$, where $t=\sin\phi$

$$=2\int\frac{2t-4}{t^2-4t+5}dt+7\int\frac{1}{t^2-4t+5}dt$$

$$=2\log\left|t^2-4t+5\right|+7\tan^{-1}(\sin\phi-2)+C$$

$$=2\log\left|\sin^2\phi-4\sin\phi+5\right|+7\tan^{-1}(\sin\phi-2)+C$$

LEVEL-IV

1. $I:\displaystyle\int\frac{dx}{a^2\sin^2x+b^2\cos^2x}=\frac{1}{ab}Tan^{-1}\left(\frac{a\tan x}{b}\right)+c$

$II:\displaystyle\int\frac{3\cos x+2\sin x}{4\sin x+5\cos x}dx=\frac{23\,x}{41}+\frac{2}{41}\log|4\sin x+5\cos x|+c$

1) only I is true
2) only II is true
3) both I and II are true
4) neither I nor II is true

2. $I:\displaystyle\int\frac{a\cos x+b\sin x}{c\cos x+d\sin x}dx=$

$\dfrac{ac+bd}{c^2+d^2}x+\dfrac{ad-bc}{c^2+d^2}\log\left|c\cos x+d\sin x\right|+k$

$II:\displaystyle\int\frac{\cos x+\sin x}{\sin x-\cos x}dx=\log|\sin x-\cos x|+k$

Which of the following is true

1) Only I 2) Only II
3) Both I and II 4) neither I nor II is true

3. **Match the following:**

I. $\displaystyle\int\frac{1}{1+\cos x}dx=$ a) tan x + sec x + c

II. $\displaystyle\int\frac{1}{1-\cos x}dx=$ b) tan x - sec x + c

III. $\displaystyle\int\frac{1}{1-\sin x}dx=$ c) - cot x + cosec x + c

IV. $\displaystyle\int\frac{1}{1+\sin x}dx=$ d) - cot x - cosec x + c

1) a, b, c, d 2) b, c, a, d
3) d, c, b, a 4) c, d, a, b

4. **Match the following:**

I. $\displaystyle\int\frac{1}{\sqrt{x}}\sin\sqrt{x}\,dx=$ a) $2\ e^{\sqrt{x}}+c$

II $\displaystyle\int\frac{1}{\sqrt{x}}\cos\sqrt{x}\,dx=$ b) $2\sin\sqrt{x}+c$

III. $\displaystyle\int\frac{1}{\sqrt{x}}e^{\sqrt{x}}\,dx=$ c) $-2\cos\sqrt{x}+c$

1) a, b, c 2) b, c, a
3) c, b, a 4) c, a, b

5. Match the following.

I. $\int \frac{1}{\sqrt{4-x^2}}dx =$ a) $\sinh^{-1}\frac{x}{2}+c$

II. $\int \frac{1}{\sqrt{4+x^2}}dx =$ b) $Cosh^{-1}\frac{x}{2}+c$

III. $\int \frac{1}{\sqrt{x^2-4}}dx =$ c) $Sin^{-1}\frac{x}{2}+c$

	I	II	III
1)	a	b	c
2)	b	c	a
3)	c	a	b
4)	a	c	b

6. Match the following.

I $\int\sqrt{25-x^2}dx =$ a) $\frac{x}{2}\sqrt{25+x^2}+\frac{25}{2}\sinh^{-1}\frac{x}{5}+c$

II. $\int\sqrt{25+x^2}dx =$ b) $\frac{x}{2}\sqrt{25-x^2}+\frac{25}{2}Sin^{-1}\frac{x}{5}+c$

III. $\int\sqrt{x^2-25}dx =$ c) $\frac{x}{2}\sqrt{x^2-25}-\frac{25}{2}Cosh^{-1}\frac{x}{5}+c$

	I	II	III
1)	a	b	c
2)	b	c	a
3)	c	a	b
4)	b	a	c

7. $I:\int\frac{1}{3+4\cos x}dx=\frac{1}{\sqrt{7}}\log\left[\frac{\sqrt{7}+\tan(x/2)}{\sqrt{7}-\tan(x/2)}\right]+c$

$II:\int\frac{dx}{\cos x-\sin x}=\frac{1}{\sqrt{2}}\log\left|\tan\left(\frac{x}{2}+\frac{3\pi}{8}\right)\right|+c$

1) only I is true
2) only II is true
3) both I and II are true
4) neither I nor II is true

8. **A:** $\int\frac{3x+7}{3x^2+14x-5}dx=\frac{1}{2}\log\left|3x^2+14x-5\right|+c$

R: $\int\frac{f'(x)}{f(x)}dx=\log|f(x)|+C$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

9. **A:** $\int\tan^4 x\sec^2 x dx=\frac{1}{5}\tan^5 x+c$

R: $\int\left[f(x)\right]^n f'(x)dx=\frac{\left[f(x)^{n+1}\right]}{n+1}+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

10. **A :** $\int x^3\sin x^4 dx=-\frac{1}{4}\cos x^4+c$

R : $\int f\left(x^n\right)x^{n-1}dx=\frac{1}{n}\int f(t)dt$ **where** $t=x^n$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

11. **A :** $\int\frac{\sin(\log x)}{x}dx=-\cos\log x+c$

R : $\int F\left[f(x)\right]f'(x)dx=\int F(t)dt+c$

where $f(x)=t$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

12. **A :** $\int\frac{1}{\sqrt{x^2+2x+10}}dx=\sinh^{-1}\frac{x+1}{3}$

R: If $R:a>0,\ b^2-4ac>0$**, then**

$\int\frac{dx}{\sqrt{ax^2+bx+c}}=\frac{1}{\sqrt{a}}\sinh^{-1}\left(\frac{2ax+b}{\sqrt{4ac-b^2}}\right)$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

13. **A :** $\int \frac{1}{\sqrt{x^2-4x-5}}dx = \cosh^{-1}\frac{x-2}{3}+c$

R: If $a>0,\ b^2-4ac>0$**, then**

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{a}}\cosh^{-1}\left(\frac{2ax+b}{\sqrt{b^2-4ac}}\right)$$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

14. **A :** $\int \frac{1}{\sqrt{2x-3x^2+1}}dx = \frac{1}{\sqrt{3}}\sin^{-1}\frac{3x-1}{2}+c$

R: If $a<0,\ b^2-4ac>0$**, then**

$$\int \frac{dx}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{-a}}\sin^{-1}\left(\frac{-2ax-b}{\sqrt{b^2-4ac}}\right)$$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

15. **A :** $\int \frac{2\cos+3\sin x}{4\cos x+5\sin x}dx = -\frac{2}{41}$

$\log|4\cos x+5\sin x|+\frac{23}{41}x+c$

R : $\int \frac{a\cos x+b\sin x}{c\cos x+d\sin x}dx = \frac{ac+bd}{c^2+d^2}x+\frac{ad-bc}{c^2+d^2}$

$\log|c\cos x+d\sin x|+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

16. **A :** $\int \frac{1}{3+4\cos x}dx = \frac{1}{\sqrt{7}}\log\left|\frac{\sqrt{7}+\tan(x/2)}{\sqrt{7}-\tan(x/2)}\right|+c$

R: If $a<b$**, then** $\int \frac{dx}{a+b\cos x} = \frac{1}{\sqrt{b^2-a^2}}$

$\log\left|\frac{\sqrt{b+a}+\sqrt{b-a}\tan(x/2)}{\sqrt{b+a}-\sqrt{b-a}\tan(x/2)}\right|+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

17. **A :** $\int \frac{1}{3+2\cos x}dx = \frac{2}{\sqrt{5}}\tan^{-1}\left(\frac{1}{\sqrt{5}}\tan\frac{x}{2}\right)+c$

R: If $a>b$**, then** $\int \frac{dx}{a+b\cos x} = \frac{2}{\sqrt{a^2-b^2}}$

$\tan^{-1}\left[\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}\right]+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

18.

A : $\int \frac{1}{5+4\sin x}dx = \frac{2}{3}\tan^{-1}\left(\frac{4+5\tan(x/2)}{3}\right)+c$

R: if $a>0,\ a>b$**, thn** $\int \frac{dx}{a+b\sin x} = \frac{2}{\sqrt{a^2-b^2}}$

$\tan^{-1}\left[\frac{b+a\tan(x/2)}{\sqrt{a^2-b^2}}\right]+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

19. **A :** $\int \frac{1}{4+5\sin x}dx = \frac{1}{3}\log\left|\frac{2\tan(x/2)+1}{2\tan(x/2)+4}\right|+c$

R: If $0<a<b$, **then** $\int \frac{dx}{a+b\sin x} = \frac{1}{\sqrt{b^2-a^2}}$

$\log\left|\frac{a\tan(x/2)+b-\sqrt{b^2-a^2}}{a\tan(x/2)+b+\sqrt{b^2-a^2}}\right|+c$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

20. **A :** $\int e^x\left(\frac{1+x\log x}{x}\right) = e^x\log x + c$

R : $\int e^x\left[f(x)+f'(x)\right]dx = e^x\log x + c$
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

21. **A :** $\int e^{5x}\sin 6x dx = \frac{e^{5x}}{61}(5\sin 6x - 6\cos 6x)+c$

R : $\int e^{ax}\sin(bx+c)dx = \frac{e^{ax}}{a+b^2}$

$\left[a\sin(bx+c)-b\cos(bx+c)\right]+c$
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

22. **A :** $\int e^{3x}\cos 4x dx = \frac{e^{3x}}{25}(3\cos 4x + 4\sin 4x)+c$

R : $\int e^{ax}\cos(bx+c)dx = \frac{e^{ax}}{a^2+b^2}$

$\left[a\cos(bx+c)+b\sin(bx+c)\right]+c$
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

23. **Observe the following statements :**

A : $\int\left(\frac{x^2-1}{x^2}\right)e^{\frac{x^2+1}{x}}dx = e^{\frac{x^2+1}{x}}+c$

R : $\int f'(x)e^{f(x)}dx = e^{f(x)}+c$.

Then which of the following is true
1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false 4) A is false but R is true

24. **Let** $I_{m,n} = \int \sin^m x.\cos^n x dx, m,n \in$ **N and also** $m,n \geq 2$

Statement I : $I_{m,n} = \frac{\sin^{m+1}x.\cos^{n-1}x}{m+n}+\frac{n-1}{m+n}I_{m,n-2}$

Statement II:

$I_{m,n} = \frac{-\sin^{m-1}x.\cos^{n+1}x}{m+n}+\frac{m-1}{m+n}I_{m-2,n}$

The incorrect statement is
1) only I 2) only II
3) both I and II 4) neither I nor II

LEVEL - IV - KEY

1) 3	2) 3	3) 4	4)3	5) 3	6) 4
7) 3	8) 1	9) 1	10) 1	11) 1	12) 1
13) 1	14) 1	15) 1	16) 1	17) 1	18) 1
19) 1	20) 1	21) 1	22) 1	23) 1	24) 4

LEVEL - IV - HINTS

1. I : Divide nr and dr with $\sec^2 x$

II : $\int \frac{a\cos x + b\sin x}{c\cos x + d\sin x}dx =$

$\left(\frac{ac+bd}{c^2+d^2}\right)x+\left(\frac{ad-bc}{c^2+d^2}\right)\log|c\cos x + d\sin x|+k$

2. I : Conceptual II : $\int \frac{a\cos x + b\sin x}{c\cos x + d\sin x}dx =$

$\left(\frac{ac+bd}{c^2+d^2}\right)x+\left(\frac{ad-bc}{c^2+d^2}\right)\log|c\cos x + d\sin x|+k$

3. I : Multiply and divide with (1- cosx)
II : Multiply and divide with (1+cosx)
III : Multiply and divide with (1+sinx)
IV : Multiply and divide with (1- sinx)

4. I : Put $\sqrt{x}=t$ II : Put $\sqrt{x}=t$ III: Put $\sqrt{x}=t$
5&6. Standard formulae

7. I : Put $\tan\frac{x}{2}=t$ II : Put $\tan\frac{x}{2}=t$
8 to 24 are standard formulae based